

Factoring Special Forms: Squares

Answer Key

Algebra 1 Polynomials

Quick Answers

1. $(x - 7)(x + 7)$

2. $(x + 5)^2$

3. $(3x - 4)(3x + 4)$

4. $(2x - 3)^2$

5. $(x - 4)(x + 4)$

6. $(x + 5)(2x - 10)$

7. $(x + 7)^2$

8. $(x - 3)(x + 3)$, $x^2 - 6x + 9$

9. $(3x + 4)(6x - 8)$

10. $(x - 3)(x + 3)(x^2 + 9)$

1. Factor $x^2 - 49$.

Step 1. Two terms with a minus sign between them: test the difference of squares. $x^2 = (x)^2$ and $49 = (7)^2$, so $A = x$ and $B = 7$.

Step 2. $A^2 - B^2 = (A - B)(A + B)$ gives $(x - 7)(x + 7)$.

Step 3. Check by expanding: the two middle terms $+7x$ and $-7x$ cancel, leaving $x^2 - 49$
 $\checkmark$ $(x - 7)(x + 7)$

2. Factor $x^2 + 10x + 25$.

Step 1. Three terms, first and last both squares: $x^2 = (x)^2$, $25 = (5)^2$. Test the perfect-square pattern.

Step 2. The pattern needs a middle term of $2 \cdot x \cdot 5 = 10x$, and that is exactly what is there.

Step 3. Because the middle sign is positive, the factor is a sum: $(x + 5)^2$. $(x + 5)^2$

3. Factor $9x^2 - 16$.

Step 1. $9x^2$ is a square as long as the coefficient is: $9x^2 = (3x)^2$, and $16 = (4)^2$, so $A = 3x$ and $B = 4$.

Step 2. $(3x)^2 - (4)^2 = (3x - 4)(3x + 4)$.

Step 3. Expanding gives $9x^2 + 12x - 12x - 16$, and the middle terms cancel $\checkmark$

$(3x - 4)(3x + 4)$

4. Factor $4x^2 - 12x + 9$.

Step 1. $4x^2 = (2x)^2$ and $9 = (3)^2$, so if this is a perfect square the roots are $A = 2x$ and $B = 3$.

Step 2. Test the middle term: $2AB = 2 \cdot 2x \cdot 3 = 12x$, and the trinomial has $-12x$. The sizes match and the sign is negative, so the factor is a difference.

Step 3. $4x^2 - 12x + 9 = (2x - 3)^2$.

$(2x - 3)^2$

5. Patio of side x with a 4 by 4 hot tub removed.

Step 1. The paved region is the big square minus the small square: $x^2 - 16$, which is $(x)^2 - (4)^2$.

Step 2. Difference of squares: $(x - 4)(x + 4)$.

Step 3. The factors are lengths in metres, and they are meaningful: cut the paved region into two rectangles and rearrange, and it measures $x - 4$ by $x + 4$. The area is unchanged, which is the geometric reason the identity is true.

$$(x - 4)(x + 4)$$

6. Factor $2x^2 - 50$.

Step 1. $2x^2$ is not a perfect square, so the difference-of-squares pattern does not apply yet. Both terms share a factor of 2, so take it out first.

Step 2. $2x^2 - 50 = 2(x^2 - 25)$.

Step 3. Now the bracket is a difference of squares: $x^2 - 25 = (x - 5)(x + 5)$, so the complete factorization is $2(x - 5)(x + 5)$.

$$2(x - 5)(x + 5)$$

7. Square mosaic of area $x^2 + 14x + 49$.

Step 1. A square's area is $(\text{side})^2$, so factoring the area as a perfect square is what produces the side length.

Step 2. $x^2 = (x)^2$, $49 = (7)^2$, and the middle term must be $2 \cdot x \cdot 7 = 14x$ — it is.

Step 3. The area is $(x + 7)^2$, so the side of the mosaic measures $x + 7$ centimetres.

$$(x + 7)^2$$

8. Error analysis: is $x^2 - 9 = (x - 3)^2$?

Step 1. Expand the student's answer: $(x - 3)^2 = x^2 - 3x - 3x + 9 = x^2 - 6x + 9$.

Step 2. That is not $x^2 - 9$: it has an extra $-6x$ and the constant has the wrong sign. So the student's factorization is wrong.

Step 3. $x^2 - 9$ has two terms, not three, so the pattern it fits is the difference of squares: $x^2 - 9 = (x - 3)(x + 3)$.

Manual review: the written sentence is an open response. Accept any wording saying the student used the perfect-square pattern on a two-term expression, or that squaring a binomial produces a middle term the original never had.

$$x^2 - 6x + 9 \quad \text{and} \quad (x - 3)(x + 3)$$

9. Factor $18x^2 - 32$.

Step 1. Neither $18x^2$ nor 32 is a perfect square, but both are even: the greatest common factor is 2.

Step 2. $18x^2 - 32 = 2(9x^2 - 16)$, and now the bracket is a difference of squares with $A = 3x$, $B = 4$.

Step 3. $2(3x - 4)(3x + 4)$. Expanding back gives $2(9x^2 - 16) = 18x^2 - 32$ ✓

$$2(3x - 4)(3x + 4)$$

10. Factor $x^4 - 81$.

Step 1. $x^4 = (x^2)^2$ and $81 = (9)^2$, so this is a difference of squares with $A = x^2$ and $B = 9$.

Step 2. $x^4 - 81 = (x^2 - 9)(x^2 + 9)$.

Step 3. The first bracket is itself a difference of squares: $x^2 - 9 = (x - 3)(x + 3)$. The second is a *sum* of squares, which does not factor over the integers, so the factorization stops here.

$$(x - 3)(x + 3)(x^2 + 9)$$